

Homework 1

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October 4, 2026

Problem 1

Problem Statement

Let a and b be integers, Prove that if $2 + a$ and $35 - b$ are divisible by 11, then $a + b$ is divisible by 11.

Solution

Proof. Since $2 + a$ and $35 - b$ are divisible by 11, it can be represented with the following equations:

$$\begin{aligned}2 + a &= 11 \cdot p, & p &\in \mathbb{Z} \\35 - b &= 11 \cdot q, & q &\in \mathbb{Z}\end{aligned}$$

where p and q are integers.

Each equations can be rearranged as follows.

$$\begin{aligned}a &= 11 \cdot p - 2 \\b &= 35 - 11 \cdot q\end{aligned}$$

The sum $a + b$ is then as follows.

$$\begin{aligned}a + b &= 11p - 2 + 35 - 11q \\&= 11 \cdot (p - q) + 33 \\&= 11 \cdot (p - q) + 11 \cdot 3 \\&= 11 \cdot (p - q + 3) \\&= 11 \cdot k & k = p - q + 3 \in \mathbb{Z}\end{aligned}$$

k is guaranteed to be an integer since it is a sum of integers. Since $a + b$ is a product of 11 and some integer k , $a + b$ is divisible by 11.

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Problem 2

Textbook p15 (15.c, 15.e)

Problem Statement

Use Euclidean algorithm to find:

(a) $(1003, 456)$

(b) $(5858, 1436)$

Solution

(a)

$$\begin{aligned}(1003, 456) &= (456, 91) \\ &= (91, 1) \\ &= (1, 0) \\ &= 1\end{aligned}$$

$$\begin{aligned}1003 &= 456 \times 2 + 91 \\ 456 &= 91 \times 5 + 1 \\ 91 &= 1 \times 91 + 0\end{aligned}$$

Answer: $(1003, 456) = 1$

(b)

$$\begin{aligned}(5858, 1436) &= (1436, 114) \\ &= (114, 68)\end{aligned}$$

$$\begin{aligned}5858 &= 1436 \times 4 + 114 \\ 1436 &= 114 \times 12 + 68\end{aligned}$$

Problem 3

Problem 4

Problem 5

Problem 6

Problem 7